📄 MILESTONE-3 DOCUMENTATION

Project Title: AI-Powered-Enhanced EHR Imaging & Documentation System
Team Name & Members: Team A

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◈ Project Vision

Healthcare professionals often spend their time on documentation and administrative tasks.

Our project aims to reduce this burden by using **Generative AI** to:

1. Generate structured **clinical notes** from raw patient data.
2. Automate **ICD-10 coding** for diagnoses.
3. Build an **end-to-end prototype** integrating these tasks seamlessly.

◈ Connection with Milestones

- Milestone 1: Data Collection & Preprocessing
    - Gathered imaging + clinical datasets (EHR, ICD-10).

    - Cleaned, standardized, and prepared datasets for GenAI compatibility.
- Milestone 2: Data-to-Insights
    - Developed methods to turn structured data into clinical notes.
    - Automated ICD-10 mapping (lookup + AI-driven).
- Milestone 3: Automation Prototype (This Module)
    - Combined all steps into a working pipeline from raw data → AI note → ICD-10 code → final output.

◈ Milestone Progression

Milestone 1: Data Collection & Preprocessing

- Collected medical imaging datasets (CT, MRI, X-ray, Ultrasound).
- Gathered structured and unstructured EHR data (age, gender, diagnosis, ICD codes).
- Cleaned, standardized, and labeled data.
- Ensured privacy by using **synthetic or open-source datasets**.

Challenge: Ensuring data quality while dealing with unstructured formats and duplicates.

Milestone 2: Data-to-Insights

- Converted structured patient attributes into **clinical documentation format**.
- Designed **lookup and AI-driven mapping** for ICD-10 coding.
- Developed logic to handle **multiple diagnoses per patient**.

Challenge: Maintaining accuracy in ICD-10 mapping and consistency across different cases.

Milestone 3: Automation Prototype (This Module)

Step 1: Prepare Sample Patient Data

- Since real EHR cannot be used, we created **10+ synthetic cases**.
- Stored as CSV/JSON for reproducibility.

Example:

Age	Gender	Symptoms	Diagnosis
45	Male	Fever, Coughness, Shortness of breath	Pneumonia

Step 2: Clinical Note Generation

- Used **Azure OpenAI (GPT-4/3.5)** to generate professional notes.
- Notes follow **SOAP format** (Subjective, Objective, Assessment, Plan).

Example AI Output:

- **Patient:** 45-year-old male
- **Presenting Complaints:** Fever, cough, shortness of breath
- **Assessment:** Likely pneumonia
- **Plan:** Chest X-ray, antibiotics

Challenge: Ensuring notes are **consistent and medically accurate** without hallucinations.

Step 3: ICD-10 Coding Automation

- **Option 1 (Lookup):** CSV file maps conditions → codes.
- **Option 2 (AI-driven):** Ask GPT directly for ICD-10.

Example:

- Pneumonia → **J18.9**
- Hypertension → **I10**

Challenge: Lookup tables need constant updating; AI sometimes gives multiple/ambiguous codes.

Step 4: Integration (Prototype)

- Python script connects everything:
    1. Reads patient data.
    2. Calls AI → generates note.
    3. Maps ICD-10 code.
    4. Saves results to output.csv.

Output Example:

Patient	Symptoms	Note generated	ICD-10
1	Fever, Cough	Patient shows signs of pneumonia..	J18.9

Step 5: Documentation & Submission

- Input files: CSV/JSON of patient records.
- Code: Python + Azure OpenAI integration.
- Output: Notes + ICD-10 codes in CSV/JSON.

- README.md with explanation of workflow, assumptions, and limitations.

◈ Progress So Far

- ✅ Dummy patient dataset created.
- ✅ Notes successfully generated with Azure GPT.
- ✅ ICD-10 mapping works (lookup + AI).
- ✅ Prototype tested and outputs verified.

◈ Known Issues

1. Limited **ICD-10 coverage** (our lookup table has only common conditions).
2. Handling **multi-diagnosis patients** is tricky (AI sometimes misses secondary codes).
3. AI-generated notes sometimes vary in **length and format**.
4. **Azure API dependency** (internet, API key).

◈ How to Reproduce

1. Prepare patient_data.csv with fields: Age, Gender, Symptoms, Diagnosis.
2. Run clinical_pipeline.py.
3. Output saved as results/output_notes.csv with:
    - Patient Data
    - AI-Generated Clinical Note

    - ICD-10 Code

◈ Real-World Impact

- Doctors spend less time on manual documentation.
- Hospitals get **standardized notes** and **accurate coding**.
- Lays foundation for **multi-modal integration** (images + EHR + coding).

◈ Team Contributions

- **Data Preparation (Milestone 1):** Team collected & cleaned imaging + EHR data.
- **Coding (Milestone 2):** Team designed lookup + AI ICD mapping.
- **Prototype Integration (Milestone 3):** Team built Python script + outputs.
- **Documentation:** Prepared detailed workflow, challenges, and progress report.

✨ Final Summary

This documentation shows our journey from **Milestone 1 (Data Collection)** → **Milestone 2 (Data-to-Insights)** → **Milestone 3 (Automation Prototype)**.

We have successfully:

- Built sample patient datasets.
- Automated note generation with GenAI.
- Mapped diagnoses to ICD-10 codes.
- Integrated everything into a working pipeline.

👉 The project demonstrates not only technical skills but also teamwork, problem-solving, and a strong foundation for future **AI-enabled healthcare solutions**.